Infraestructures del Transport Aeri

WP1 Simulator Development

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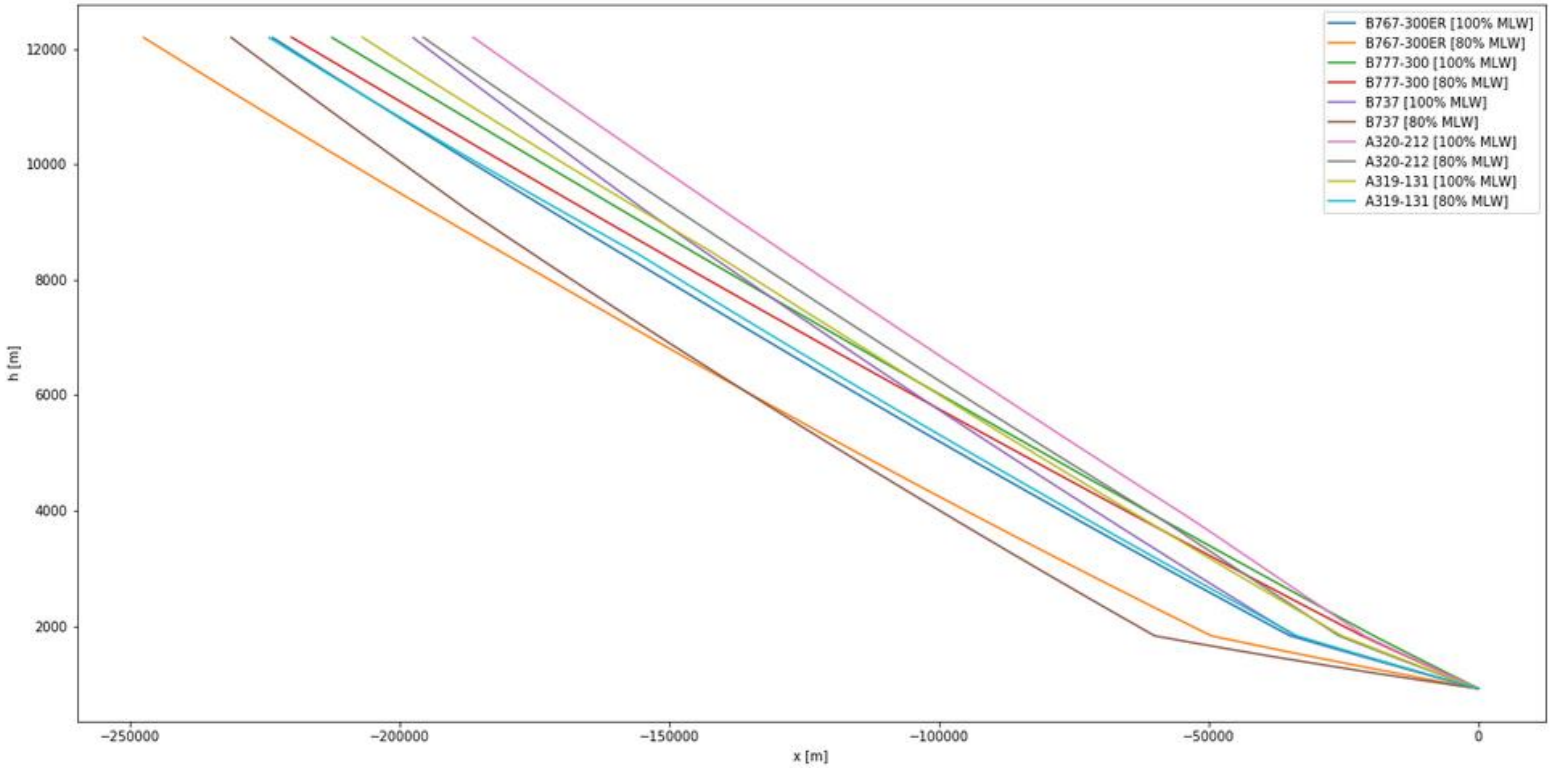
Goal of the simulator

Simulate descending trajectories in a Continuous Descent Operation (CDO)

The simulator must allow to know the altitude at which is flying an aircraft at a certain distance from the Initial Approach Fix (IAF, x=0)



Assumptions

Speed

- Speed that minimises the **rate of descent**

Session 1 results

Atmosphere

- The atmosphere will follow the **International Standard Atmosphere (ISA)**

Search ISA formulation

Thrust

- Engine **thrust** is assumed always to be **idle**

$$T_{max} = C_{T1}\left(1 - \frac{h_p}{C_{T2}} + C_{T3}{h_p}^2\right)$$

$$T_{desc} = C_{Tdesc} \cdot T_{max}$$

Assumptions

Performance models

- In order to model aircraft performance the EUROCONTROL's base of aircraft data (BADA) shall be used

$$CD = CD_0 + CD_2 CL^2$$

$$\eta = C_{F1}\left(1 + \frac{v}{C_{F2}}\right)$$

$$FF = \eta T$$

CD_0 and CD_2 change as a function of the altitude in order to simulate the change in the flap configuration

Assumptions

Parameters for the aircraft

	B767-300ER	B777-300	B737	A320-212	A319-131
Max Landing Weight (tons)	0.145150E+03	0.237680E+03	0.51710E+02	0.64500E+02	0.61000E2
Max Weight (tons)	0.20410E+03	0.29930E+03	0.70800E+02	0.77000E+02	0.70000E+02
Max. Payload (tons)	0.46500E+02	0.64900E+02	0.16920E+02	0.21500E+02	0.17000E+02
S (m^2)	0.28350E+03	0.42804E+03	0.12465E+03	0.12260E+03	0.12260E+03
$CD_{0,\,app}$	0.14000E-01	0.17300E-01	0.27000E-01	0.24200E-01	0.28400E-01
$CD_{2,\,app}$	0.49000E-01	0.48400E-01	0.44100E-01	0.46900E-01	0.37600E-01
$CD_{0.\,clean}$	0.17400E-01	0.15700E-01	0.23500E-01	0.24000E-01	0.28000E-01
$CD_{2,\,clean}$	0.45900E-01	0.42000E-01	0.44500E-01	0.37500E-01	0.31000E-01
$h_{p,desc}$ (ft)	26418	36122	30152	12398	27726
$CT_{Desc,high}$	0.64359E-1	0.44239E-1	0.36336E-1	0.45711E-1	0.83084E-1
$CT_{Desc,low}$	0.55988E-1	0.41065E-1	0.53395E-1	0.27207E-1	0.51765E-1
$CT_{Desc,app}$	0.12475	0.92921E-1	0.16440	0.13981	0.14767
C_{T1} (N)	.35167E+06	.42577E+06	.14573E+06	.13605E+06	.13900E+06
C_{T2} (ft)	.44673E+05	.48987E+05	.55638E+05	.52238E+05	.58900E+05
C_{T3} ($1/ft^2$)	.10129E-09	.66146E-10	.14200E-10	.26637E-10	.57200E-14
C_{F1} (kg/(min·kN))	.54005E+00	.87843E+00	.94680E+00	.94000E+00	.68800E+00
C_{F2} (kt)	.55782E+03	.36897E+04	.10000E+15	.10000E+06	.16700E+04

Basic "How to"

How to build the simulator

1. **WHAT SHOULD MY CODE DO**?

Your code has to **generate a 2D trajectory [x,h]** by calling a **FUNCTION** from the main script with some **input parameters** (**final** weight, aircraft model)

We cannot use the weight at the Top of Descent (TOD) as an initial condition because we don't know it. To overcome this issue, **we will set the final weight as a percentage of the MLW (80% or 100%)** and **the trajectory will be simulated backwards** (from the IAF to Hmax=FL400)

Basic "How to"

How to build the simulator

2. HOW CAN I ACHIEVE THIS?

Knowing the initial position you should implement a **FUNCTION** containing an **iterative process (while / for loop)** in which all points of the trajectory shall be computed. **At each iteration, one point is computed (x,h)**.

Since the trajectory will be simulated backwards, the initial position will actually be the last position (IAF → x = 0, h = 5000 ft).

- You shall consider that an iteration lasts for **1 second** ("delta_t")
- You shall assume that the speed remains constant within each iteration

Basic "How to"

What is a FUNCTION

*"A function is a block of organized, reusable code that is used to perform a single, related action. Functions provide better modularity for your application and a high degree of **code reusing**"* (definition extracted from **tutorialspoint**)

<u>It is mandatory</u> to implement a function like this (MATLAB nomenclature):

```
function [x,h] = getCDO(aircraft_model,MLW_percent)
```

- Inside the function you have to put all the code required to compute the trajectory points.

- This function takes as input two parameters (aircraft_model and MLW_percent) and returns a trajectory defined by two vectors (x,h). **The function can be called as many times as required with as many combinations of input parameters as required from the main script**. Example (MATLAB): `[x,h] = getCDO('B767',100)`

Basic “How to”

Your simulator SHOULD

- Generate all the trajectories with a single “click”
- Allow you to use all computed trajectories for future tasks (e.g. plot all trajectories in a chart, use the trajectories in further analyses, etc.)

Your simulator SHOULD NOT

- Ask you to type any input parameter once the execution has started

WP1 organisation

16 September	KO + Speed minimum RoD	Literature review
23 September	Simulator (I)	
30 September	Simulator (II)	
7* October	Simulator (III) + Sequencing	
14 October	Sequencing + STAR altitude constraints + Report draft review	

WP1 report deadline -> **26th October at 23:55**

Thank you!!

Gràcies!!